

# Comparative study of air quality data of Senegal(Dakar) and Cameroon(Douala)

Group 5

African Institute for Mathematical Sciences, AIMS-Senegal

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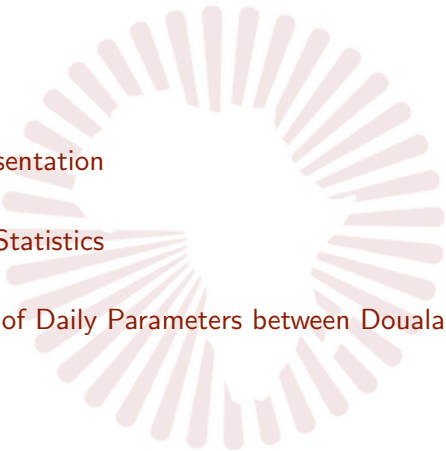
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# List of Students in Project

- Djibril Ndour[data cleaning, search for parameter thresholds, Recommendation and beamer]
- Farouk Zainab[Correlation analysis Data cleaning]
- Fatou Diop[ beamer presentation]
- Bayiha Jean[Analysis and visualisation of Douala Data]
- Hounye DAMIEN ZINSOU[Analysis and visualisation of Dakar Data]



# Overview

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- 1 Introduction
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  - 5 Conclusion

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# Introduction

The manipulation of data is a key step in transforming raw information into actionable knowledge. For our project, we downloaded data from the OpenAQ platform to analyze and compare air quality in two African countries: Senegal and Cameroon. We chose to focus on the cities of Dakar and Douala, the main metropolises of these countries. The data for Dakar is recorded in the file *air-qualdakarsept\_oct.csv*, while the data for Douala is found in the file *air\_qualdoualasept.csv*. By linking and analyzing these data, we aim to obtain relevant information on air quality variations in these two regions and to identify the factors that influence these variations.

**As shown in the following photos**

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# Introduction

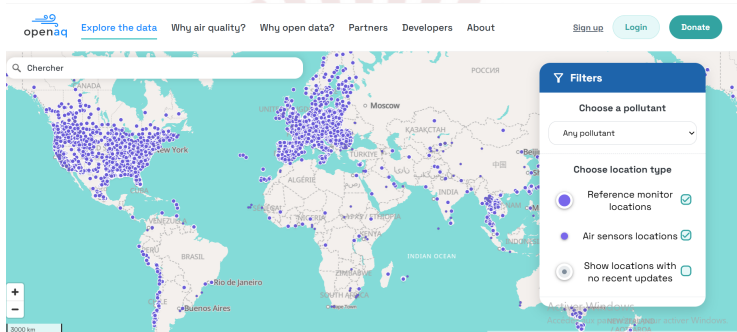


Figure: Air quality monitoring stations

# Introduction

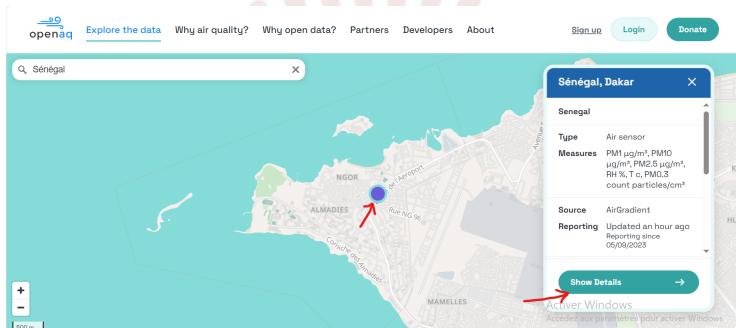


Figure: Air quality monitoring stations for Dakar

# Introduction

## Download

### Download Data CSV

Data downloads on OpenAQ Explorer are limited to 1000 records per parameter or 30 days, whichever is larger. For larger downloads use the OpenAQ API.

Read more about the OpenAQ API at <https://docs.openaq.org>

Start date

28/10/2024



End date

29/10/2024



PM1  $\mu\text{g}/\text{m}^3$



PM10  $\mu\text{g}/\text{m}^3$



PM2.5  $\mu\text{g}/\text{m}^3$



RH %



T c



PM0.3 count particles/cm<sup>3</sup>



Download CSV



Activier Windows

Accédez aux paramètres pour activer Windows

Figure: Part to download Data CSV



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# Introduction

- These charts illustrate the different steps to follow to access the data after clicking on the following link:  
<https://openaq.org/>.
- Similarly, we followed the same process to obtain the data from the file *douala\_openaq\_2022\_2024.csv*.





# Data of Presentation

After downloading our data during the same period from 2024-01-09 to 2024-01-10, we proceeded to display it to get an overview and the following information:

|   | location_id | location_name  | parameter | value    | unit  | datetimeUtc          | datetimeLocal        | timezone     | latitude  | longitude  | country_iso | isMobile | isMonitor |
|---|-------------|----------------|-----------|----------|-------|----------------------|----------------------|--------------|-----------|------------|-------------|----------|-----------|
| 0 | 1531944     | Sénégal, Dakar | pm1       | 8.366459 | µg/m³ | 2024-09-23T01:00:00Z | 2024-09-23T01:00:00Z | Africa/Dakar | 14.746476 | -17.510441 | NaN         | NaN      | NaN       |
| 1 | 1531944     | Sénégal, Dakar | pm1       | 8.810278 | µg/m³ | 2024-09-23T02:00:00Z | 2024-09-23T02:00:00Z | Africa/Dakar | 14.746476 | -17.510441 | NaN         | NaN      | NaN       |
| 2 | 1531944     | Sénégal, Dakar | pm1       | 8.062624 | µg/m³ | 2024-09-23T03:00:00Z | 2024-09-23T03:00:00Z | Africa/Dakar | 14.746476 | -17.510441 | NaN         | NaN      | NaN       |

Figure: raw data dakar

|      | location_id | location_name | parameter   | value     | unit  | datetimeUtc          | datetimeLocal             | timezone      | latitude | longitude | country_iso |
|------|-------------|---------------|-------------|-----------|-------|----------------------|---------------------------|---------------|----------|-----------|-------------|
| 3403 | 3075781     | COLLEGE NAL   | temperature | 28.108334 | c     | 2024-10-16T15:00:00Z | 2024-10-16T16:00:00+01:00 | Africa/Douala | 4.048471 | 9.769506  | NaN         |
| 420  | 3075781     | COLLEGE NAL   | pm1         | 1.308333  | µg/m³ | 2024-10-13T16:00:00Z | 2024-10-13T17:00:00+01:00 | Africa/Douala | 4.048471 | 9.769506  | NaN         |
| 1751 | 3075781     | COLLEGE NAL   | pm25        | 11.195833 | µg/m³ | 2024-10-08T09:00:00Z | 2024-10-08T10:00:00+01:00 | Africa/Douala | 4.048471 | 9.769506  | NaN         |

Figure: raw data Douala



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## Description of Data Columns (1/3)

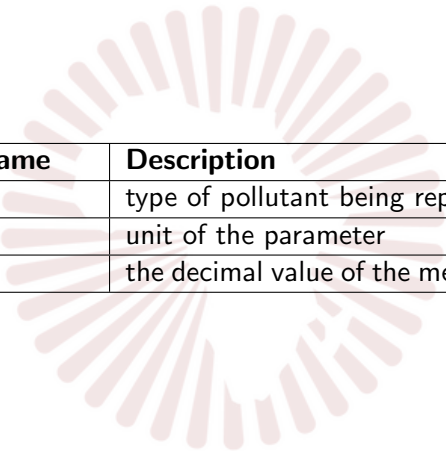
| Column name | Description                                                              |
|-------------|--------------------------------------------------------------------------|
| location_id | the OpenAQ location id of the station                                    |
| sensor_id   | the OpenAQ sensor id of the pollutant measured                           |
| location    | name of location                                                         |
| datetime    | date and time in ISO-8601 format with timezone offset of the measurement |

## Description of Data Columns (2/3)

| Column name | Description                                                                                                    |
|-------------|----------------------------------------------------------------------------------------------------------------|
| latitude    | decimal degree (EPSG:4326) representation of the latitude (Y) of the sensor. Minimum of -90. Maximum of 90.    |
| longitude   | decimal degree (EPSG:4326) representation of the longitude (X) of the sensor. Minimum of -180, Maximum of 180. |



## Description of Data Columns (3/3)



| Column name | Description                          |
|-------------|--------------------------------------|
| parameter   | type of pollutant being reported     |
| unit        | unit of the parameter                |
| value       | the decimal value of the measurement |

## Data of Presentation

At parameter level, it should be noted that the presence of pollutants being reported as :  
pm1,pm10,pm2.5,relativehumidity,temperature,um003. the display of our data shows a number of anomalies at table level. the first point to note is that the parameters we need to study are grouped together in a single column. So we're going to use the `pivot_table()` function to put each parameter component in a column with its values.

# After data processing

| parameter | datetimeLocal        | location_id | location_name  | latitude  | longitude  | pm1       | pm10      | pm25      | relativehumidity | temperature | um003       |
|-----------|----------------------|-------------|----------------|-----------|------------|-----------|-----------|-----------|------------------|-------------|-------------|
| 427       | 2024-10-26T21:00:00Z | 1531944     | Sénégal, Dakar | 14.746476 | -17.510441 | 12.028587 | 20.812915 | 19.828912 | 78.185078        | 29.939681   | 2295.615824 |
| 43        | 2024-09-24T20:00:00Z | 1531944     | Sénégal, Dakar | 14.746476 | -17.510441 | 9.462407  | 16.193913 | 15.512940 | 84.479952        | 28.336683   | 1825.470713 |
| 378       | 2024-10-11T21:00:00Z | 1531944     | Sénégal, Dakar | 14.746476 | -17.510441 | 24.988402 | 41.335538 | 38.596185 | 84.137941        | 28.680669   | 5470.477295 |

Figure: data Dakar after processing

| parameter | datetimeLocal             | location_id | location_name | latitude | longitude | pm1      | pm10      | pm25      | relativehumidity | temperature | um003      |
|-----------|---------------------------|-------------|---------------|----------|-----------|----------|-----------|-----------|------------------|-------------|------------|
| 0         | 2024-09-23T11:00:00+01:00 | 3075781     | COLLEGE NAL   | 4.048471 | 9.769506  | 6.516667 | 19.038889 | 15.127778 | 54.322222        | 26.792778   | 524.800001 |
| 1         | 2024-09-23T12:00:00+01:00 | 3075781     | COLLEGE NAL   | 4.048471 | 9.769506  | 0.583333 | 2.916667  | 2.254167  | 58.566667        | 28.182083   | 235.479165 |
| 2         | 2024-09-23T13:00:00+01:00 | 3075781     | COLLEGE NAL   | 4.048471 | 9.769506  | 0.150000 | 1.380556  | 1.130556  | 57.333333        | 28.574445   | 223.355556 |
| 3         | 2024-09-23T14:00:00+01:00 | 3075781     | COLLEGE NAL   | 4.048471 | 9.769506  | 2.279167 | 9.445833  | 8.000000  | 55.183333        | 28.263333   | 384.250001 |

Figure: data Douala after processing

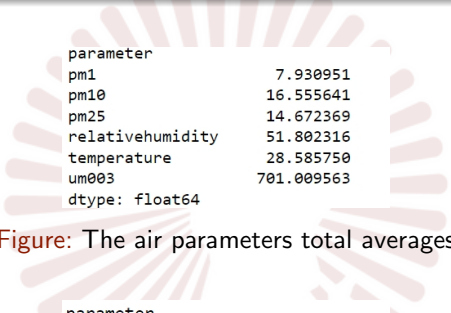


## Parameters meaning

- PM1: Particulate Matter with a diameter of 1 micrometer or less.
- PM10: Particulate Matter with a diameter of 10 micrometers or less. PM10 includes dust, pollen, mold, and other particles.
- PM2.5: Particulate Matter with a diameter of 2.5 micrometers or less.
- Relative Humidity: The amount of moisture in the air compared to the maximum amount the air can hold at a given temperature.
- Temperature: The measure of heat in the air, usually recorded in Celsius or Fahrenheit.
- UM003: Particulate Matter with a diameter of 0.3 micrometers or less.

# The air parameters total averages and median for our dataset

## For Douala



| parameter        |            |
|------------------|------------|
| pm1              | 7.930951   |
| pm10             | 16.555641  |
| pm25             | 14.672369  |
| relativehumidity | 51.802316  |
| temperature      | 28.585750  |
| um003            | 701.009563 |
| dtype:           | float64    |

Figure: The air parameters total averages

| parameter        |            |
|------------------|------------|
| pm1              | 4.845833   |
| pm10             | 11.683333  |
| pm25             | 10.194792  |
| relativehumidity | 52.400000  |
| temperature      | 28.641563  |
| um003            | 541.933334 |
| dtype:           | float64    |



# The air parameters total averages and median for our dataset

## For Dakar

```
parameter  
pm1          8.900955  
pm10         14.938158  
pm25         14.058376  
relativehumidity 85.040676  
temperature  28.411775  
um003        1893.233030  
dtype: float64
```

Figure: The air parameters total averages

```
parameter  
pm1          6.064664  
pm10         10.827274  
pm25         10.378460  
relativehumidity 85.531584  
temperature  28.245894  
um003        1289.787024  
dtype: float64
```



# Graph

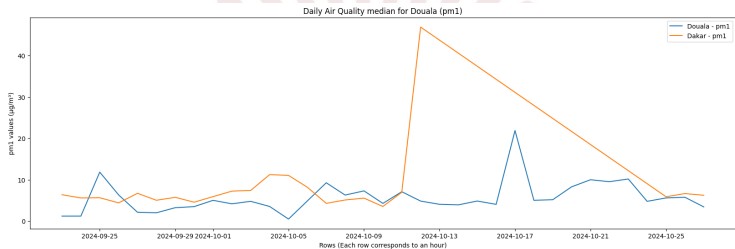


Figure: Daily Median pm1 for Douala and Dakar



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# Graph

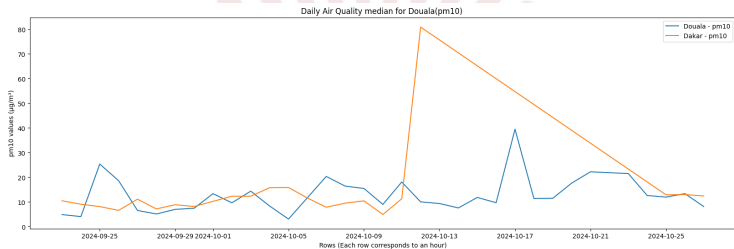


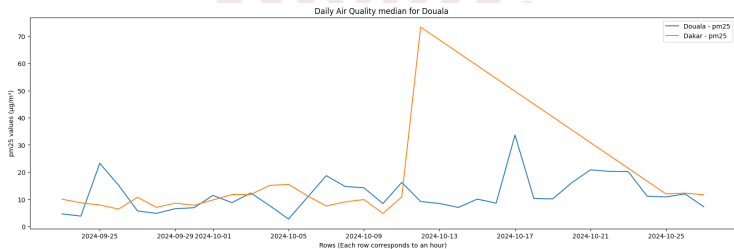
Figure: Daily Median pm10 for Douala and Dakar



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# Graph



**Figure:** Daily Median pm2.5 for Douala and Dakar



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# Graph

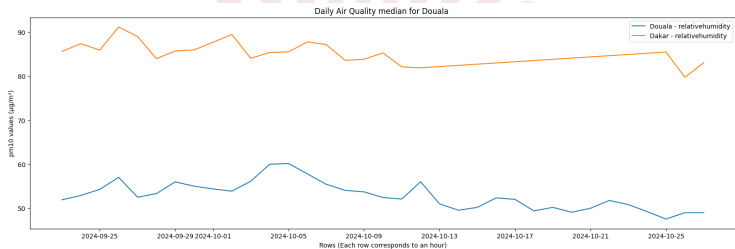


Figure: Daily Median Relative Humidity for Douala and Dakar

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# Graph

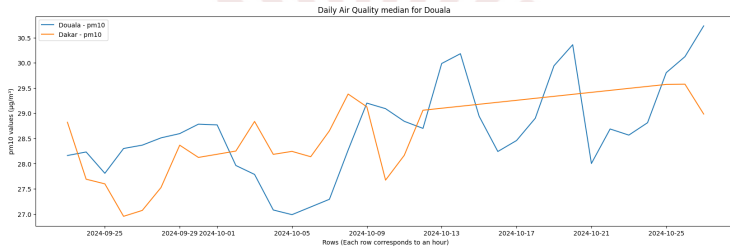


Figure: Daily Median Temperature for Douala and Dakar(pm10)

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# Graph

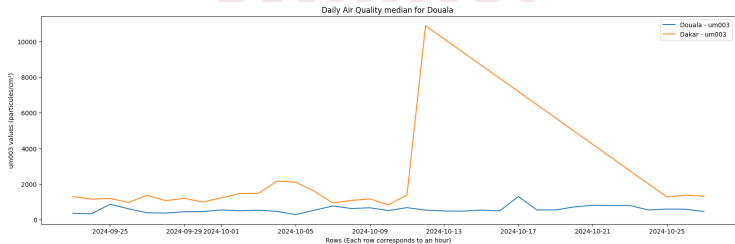
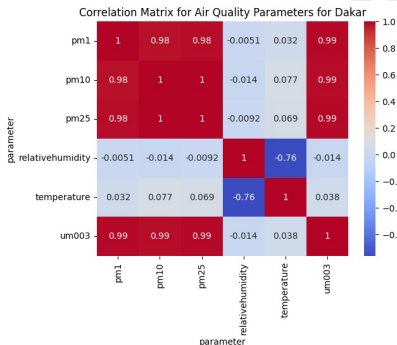


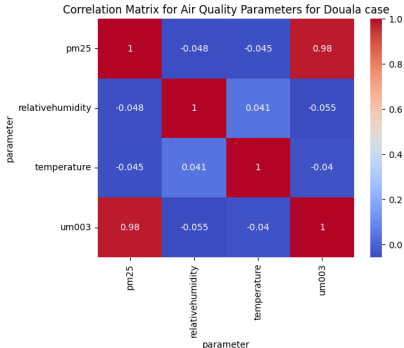
Figure: Daily Median Temperature for Douala and Dakar(um003)

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# Air Quality Parameters



correlation matrix  
air quality Dakar



correlation matrix  
air quality Douala

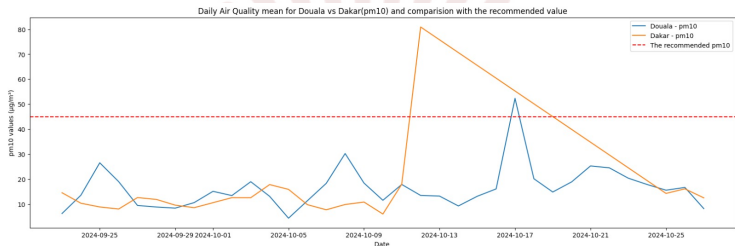


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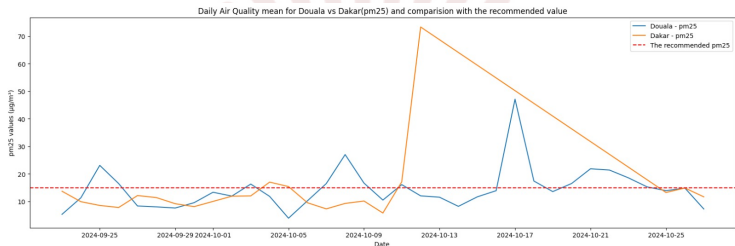
# Graph



**Figure:** Daily Air Quality mean for Douala vs Dakar (pm10) and comparison with the recommended value

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# Graph



**Figure:** Daily Air Quality mean for Douala vs Dakar (pm2.5) and comparison with the recommended value

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## Conclusion

At the end of our study, Dakar appears to be slightly more polluted in terms of ultrafine particles ( $um003$ ) and PM1, while Douala has slightly higher concentrations of PM10 and PM2.5. If pollution by ultrafine particles is considered critical for air quality, Dakar can be considered the most polluted city in this case.

Air pollution can lead to numerous illnesses, including respiratory diseases, cardiovascular diseases and other health problems. the establishment of an early warning system that will enable people to take appropriate protective measures quickly.

- promote public transport such as the TER (Regional Express Train) and the BRT (Bus Rapid Transit)
- renewing the car fleet with less polluting vehicles;
- Planting trees

# Bibliography

OpenAQ Platform for data.

the regulatory threshold table [**ecologie`normes`seuils**].

world health organization

European environmental agency



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